

Charlie Lake

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EDUCATION

Doctor of Philosophy (PhD), Biomedical Engineering

University of Toronto

Supervisor: Dr. Tom Chau

Thesis: Temporal Dynamics of Brain-Computer Interface User Skill Acquisition

Expected Graduation: June 2029

Toronto, ON

BASc, with Distinction, Biomedical Engineering

University of British Columbia (UBC)

Cumulative GPA: 93%

May 2024

Vancouver, BC

RESEARCH EXPERIENCE

Student Academic Researcher (Directed Study - BMEG 490)

UBC, Robotics and Control (RCL) Lab

Vancouver, BC

September 2023 – April 2024

- Supervisor: Dr. Tim Salcudean
- Directed an independent research project to investigate thermal Speed-of-Sound (SoS) Ultrasound imaging, for non-invasive temperature monitoring during thermal therapies
- Designed agar phantom model and data capture setup, to generate temperature differentials, capture ground truth values, and capture Ultrasound images during heating experiment
- Evaluated accuracy of existing SoS imaging methods and optimized signal processing pipeline to improve signal-to-noise ratio (SNR) and minimize temperature estimation error

Undergraduate Research Assistant

UBC, Innovation in Dementia and Aging (IDEA) Lab

Vancouver, BC

May 2021 – May 2022

- PI: Dr. Lillian Hung, work funded by **NSERC USRA**
- Led technical implementation of telepresence robots (Double 3) in long-term care homes in a Patient-Oriented Research project enabling families to connect remotely, fostering connection during the pandemic
- Developed a web app to record telepresence robot calls, enabling remote data collection during pandemic
- Designed a booking system in Python for users to schedule recurring calls on robots, facilitating family visits
- Engaged older adults living with dementia in implementation of telepresence robots in long-term care to ensure the implementation met users' individual needs

INDUSTRY EXPERIENCE

Software Engineering Co-op

IKOMED Technologies Inc

Vancouver, BC

May 2022 – April 2023

- Automated signal processing, analysis and visualization of medical device test data using Python pandas and NumPy to more quickly identify trends used in optimizing treatment parameters
- Developed in-house Raspberry Pi-based X-ray-system-emulator, with PCB developed in KiCAD, housing designed in SolidWorks, and software in Python PyQt, reducing engineering time required for system verification by 50%
- Designed a Python application to perform system diagnostics and updates on medical devices over SSH and using Docker, resulting in a 75% reduction in service errors and faster client issue troubleshooting and resolution
- Wrote clear engineering technical documentation to meet FDA medical device standards

TEACHING EXPERIENCE

Teaching Assistant (TA) APS112 (Engineering Strategies & Practice)

UofT, Faculty of Applied Science & Engineering (FASE)

Toronto, ON

January 2026 – Present

- Support 6 student design teams, providing structured feedback on technical reports and key project deliverables
- Achieved 100% favorable ratings in student feedback, reflecting strong student engagement and support

Teaching Assistant (TA) APS106 (Fundamentals of Computer Programming)

UofT, Faculty of Applied Science & Engineering (FASE)

Toronto, ON

January 2026 – Present

- Ran 4 guest lectures and 3 review lectures to classes of ~400 first year students on relevant lecture design problems
- Achieved 94% favorable ratings in student feedback, reflecting strong student engagement and support

Teaching Assistant (TA) CSC236 (Intro Theory of Computation)

Toronto, ON

UofT, Department of Computer Science

September 2025 – December 2025

- Lead weekly tutorial sessions to classes of 35 2nd-year Computer Science students on proof method topics

Computer Science (CPSC) Course Coordinator

Vancouver, BC

UBC, Department of Computer Science (CPSC)

September 2024 – August 2025

- Coordinate delivery of courses CPSC 121 and CPSC 310 at UBC, both delivered to 500+ students a term
- Manage all administrative aspects of courses, including assessment scheduling, coordination of team of >50 Teaching Assistants (TAs), and student communications, ensuring smooth course delivery

CPSC TA Training Facilitator

Vancouver, BC

UBC, Department of Computer Science (CPSC)

January 2024 – January 2025

- Facilitate and develop training sessions for start-of-term TA Training workshops held by the CPSC Department
- Sessions Run: *Assessing Students and Providing Constructive Feedback* (Jan 2024, May 2024, Jan 2025) and *Explaining Concepts & Not Giving Away Answers* (Sept 2024)

Lead TA CPSC 121

Vancouver, BC

UBC, Department of Computer Science (CPSC)

September 2020 – August 2024

- In addition to typical TA duties (below), lead weekly meetings, manage inventory of lab equipment, develop teaching resources for peer TAs, and design improved learning materials for students
- Redesigned lab curriculum with instructor guidance to improve student engagement and lab learning outcomes

Lab TA CPSC 121

Vancouver, BC

UBC, Department of Computer Science (CPSC)

September 2019 – August 2020

- Foster relationships with 100+ students per term during hands-on weekly labs about digital logic
- Achieved consistent 100% favorable ratings in student feedback, reflecting strong student engagement and support

VOLUNTEER**Discovery Physics Co-Lead**

Toronto, ON

UofT Discovery Program

September 2025 – Present

- Act as a subject lead for the physics stream of the Discovery program, an Ontario highschool curriculum-integrated STEM learning program led by UofT graduate students
- Engineered test setup for evaluation of student Water Turbine designs for use in student programming, handled student questions, and mentored new Discovery volunteers

ToBE Workshop Coordinator

Toronto, ON

Toronto Biomedical Engineering Conference (ToBE)

September 2025 – Present

- Organize two workshops including industry panel and AI ethics workshop for yearly ToBE conference

BESA University Affairs Representative

Toronto, ON

Biomedical Engineering Student Association (BESA)

September 2025 – Present

- Represent BESA in university-wide discussions and initiatives related to graduate student affairs, ensuring student perspectives are communicated and considered in institutional decision-making

CEEA Student Committee Member

Canada

Canadian Engineering Education Association (CEEA-ACÉG)

September 2025 – Present

- Contribute to the design and development of student-centered programs, including the CEEA Student Welcome Package, aimed at connecting new students with key resources and promoting student engagement within CEEA

CEEA Activities & Professional Development (A&PD) Committee Member

Canada

Canadian Engineering Education Association (CEEA-ACÉG)

September 2025 – Present

- Provide student insight on A&PD initiatives, supporting decisions related to grant allocations and contributing to the planning of professional development opportunities for CEEA members

PUBLICATIONS

L J Holanda, **C. Lake** †, F. Qureshi †, J. Gungor †, N. Youvalari †, T. Chau. 2026 (in-press). “Emerging Assistive Communication Technologies for Children and Youth with Complex Communication Needs” Current Developmental Disorders Reports.(† Equal Contribution)

Seaman et al. 2026 (in-press). “Evaluating effectiveness of an inclusive inquiry-based STEM education program on senior high school student performance post-COVID (Evaluation).” *2026 ASEE Annual Conference & Exposition*.

Charlie Lake, Alice Kang, Jess Tran. 2026 (in-press). “Reading Care Work: Imagining a Care-Centered Disability Justice Framework in Engineering Education.” *Proceedings of the Canadian Engineering Education Association (CEEAA-ACEG)*.

Jeffrey Niu, Jessica Wong, **Charlie Lake**, Justin Rahardjo, Hedayat Zarkoob, Oluwakemi Ola, Patrice Belleville, Karina Mochetti, Meghan Allen, Firas Moosvi, and Steven Wolfman. 2025. “Expanding the Horizons of Autograding: Innovative Questions at UBC.” *Proceedings of the 56th ACM Technical Symposium on Computer Science Education (SIGCSE 2025)*, February 26–March 1.

Charlie Lake, and Karina Mochetti. “Logic Circuits unveiled: Bridging the gap between discrete mathematics and computer science education.” *The 26th Western Canadian Conference on Computing Education*, vol. 74, 2 May 2024, pp. 1–7, <https://doi.org/10.1145/3660650.3660653>.

Lillian Hung, **Charlie Lake**, Ali Hussein, Joey Wong, and Jim Mann. “Using telepresence robots as a tool to engage patient and family partners in dementia research during COVID-19 pandemic: A qualitative participatory study.” *Research Involvement and Engagement*, vol. 9, no. 1, 23 Mar. 2023, <https://doi.org/10.1186/s40900-023-00421-w>.

PRESENTATIONS

Charlie Lake, Jess N. Tran, Jessica T. Wolf. “An Introduction to Anti-Deficit Theoretical Frameworks”. June 2025. Workshop: *Canadian Engineering Education Association Conference (CEEAA-ACEG)*, Montreal, Quebec, Canada.

Charlie Lake. “You can’t spell teaching without TA: An undergraduate teaching assistant’s (UTA) reflection on the benefits and challenges of UTAs in Computer Science instruction and curriculum design”. June 2024. Oral presentation: *Canadian Engineering Education Association Conference (CEEAA-ACEG)*, Edmonton, Alberta, Canada.

Charlie Lake, and Karina Mochetti. “Logic Circuits Unveiled: Bridging the Gap between Discrete Mathematics and Computer Science Education”. May 2024. Oral presentation: *Western Canadian Conference on Computing Education*, Kelowna, British Columbia, Canada.

Charlie Lake, Lillian Hung, Joey Wong, Ali Hussein, Jim Mann, and Mario Gregorio. “Using telepresence robots as a tool for virtual research during the COVID-19 pandemic”. August 2022. Poster: *Alzheimer’s Association International Conference (AAIC)*, San Diego, USA.

AWARDS

NSERC Canada Graduate Scholarship - Masters (CGSM), \$27000	2026
Kimel Family Graduate Student Scholarship in Pediatric Rehabilitation, \$20000	2025
Peterborough K.M. HUNTER Charitable Foundation Graduate Awards, \$12000	2025
Barbara & Frank Milligan Scholarship, \$7100	2025
Engineers & Geoscientists BC Achievement Award, \$200	2024

UBC Trek Excellence Scholarship, \$1500 (x3)	2019, 2021-2023
Charles & Jane Banks Scholarship, \$4500	2023
Ward-Essof Scholarship, \$1700	2023
CPSC Undergraduate TA Award, \$0	2021, 2023
NSERC Undergraduate Student Research Award (USRA), \$6000	2021
UBC J Fred Muir Memorial Scholarship in Science, \$350	2019

CERTIFICATIONS

UBC Instructional Skills Workshop (ISW) for Grad Students	August 2024
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